

Note 6.3 Arc Length

Length of a Curve $y = f(x)$

- **Definition**

If f' is continuous on $[a, b]$, then the **length(arc length)** of the curve $y = f(x)$ from the point $A = (a, f(a))$ to $B = (b, f(b))$ is the value of the integral

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- **Proof**

First, the function must be smooth and its graph must be a smooth curve, that is, f' is continuous on $[a, b]$. Then we partition the interval $[a, b]$ into n subintervals with $a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$. Then the corresponding point on the function is $P_k(x_k, y_k)$. So we approximate the curve from P_{k-1} to P_k as the straight line segment and its length is

$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

where $\Delta x_k = x_k - x_{k-1}$, $\Delta y_k = y_k - y_{k-1}$.

Then the length of the curve will be

$$\sum_{k=1}^n L_k = \sum_{k=1}^n \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

By the [Mean Value Theorem](#), there is a point $c_k \in [x_{k-1}, x_k]$ such that

$$\Delta y_k = f'(c_k) \Delta x_k$$

Thus the length of the curve will be

$$\sum_{k=1}^n L_k = \sum_{k=1}^n \sqrt{(\Delta x_k)^2 + (f'(c_k) \Delta x_k)^2} = \sum_{k=1}^n \sqrt{1 + [f'(c_k)]^2} \Delta x_k$$

Since $\sqrt{1 + [f'(c_k)]^2}$ is continuous on $[a, b]$, then when $n \rightarrow \infty$, the norm of the Riemann sum goes to zero. Thus

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n L_k = \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{1 + [f'(c_k)]^2} \Delta x_k = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

Dealing with Discontinuities in dy/dx

- **Formula for the length of $x = g(y)$, $c \leq y \leq d$**

If g' is continuous on $[c, d]$, then the **length(arc length)** of the curve $x = g(y)$ from the point $A = (g(c), c)$ to $B = (g(d), d)$ is the value of the integral

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

The Differential Formula For Arc Length

- **Definition of Arc Length Function**

By the [Fundamental Theorem Pt.1](#), we can define a new function

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} dx$$

And we can know the function is differentiable and continuous. It measures the length along the curve $y = f(x)$ from the initial point $P_0(a, f(a))$ to the point $Q(x, f(x))$.

Then the differential of arc length is

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{dx^2 + dy^2}$$

which can be integrated to give the total length of a curve, that is,

$$L = \int ds$$